Kaiwen Tian

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Education

Department of Automation, Tsinghua University *Sep. 2023 – Present*
B.Eng. in Automation (Expected Jun. 2027)

- GPA: 3.9/4.0, Rank 8/173 (Top 5%)
- Selected Coursework: Programming Fundamentals (A), Linear Algebra (A), Principles of Artificial Intelligence (A-), Pattern Recognition and Machine Learning (A-), Generative Artificial Intelligence (A-), Intelligent Speech Processing (A), Signals and System Analysis (A), Operations Research (A)

Research Interests

My previous work focused on generative models and computer vision, particularly 3D / 4D reconstruction. I am currently interested in LLMs, LLM agents, generative models, and AI for Science (AI4Science), with an emphasis on building intelligent systems for scientific research.

Publications & Preprints

[1] Sparse-View 4D Gaussian Splatting via Spatiotemporal Priors and Generative Assistance
Volumetric Video Challenge Workshop, SIGGRAPH Asia 2026 (Accepted)
🥇 **First Prize (Rank 1), Sparse-View Track** | arXiv:2610.04606
Shengqi Wang, Zhengxian Yang, **Kaiwen Tian**, Yang Liu, Bowen Liu, Hua Du, Taicheng Huang, Jiamin Wu, Tao Yu

Research Experiences

Research Assistant (LLM Agents / AI for Research) *Hong Kong, China*
Advisor: Prof. Qiang Xu, The Chinese University of Hong Kong *Sep. 2026 – Present*

- Develop LLM-based agents for scientific research workflows, focusing on the use of large language models to support AI for Science.

Research Assistant (Sparse-View Dynamic Human Reconstruction) *Beijing, China*
Advisor: Prof. Tao Yu, Tsinghua University *Oct. 2025 – Sep. 2026*

- Reproduced and benchmarked representative 3D / 4D reconstruction methods, including 3DGS, VGGT, and Diffuman4D, under sparse-view settings.
- Investigated generative-model-assisted dynamic human 4D reconstruction by conditioning the Wan video generation model with geometric priors for novel view synthesis.

Research Assistant (AI-Enhanced Autonomous Driving Simulation) *Beijing, China*
Advisor: Prof. Li Li, Department of Automation, Tsinghua University *Apr. 2025 – Jun. 2025*

- Investigated the application of large language models and deep learning to autonomous driving simulation, focusing on scenario understanding and decision planning.
- Designed and implemented evaluation metrics for assessing the simulation platform.

Internship Experience

Research Intern (LLM Agents / AI for Science) *Beijing, China*
PhAI Labs *Sep. 2026 – Present*

- Work on LLM agents and AI for Science, exploring intelligent systems for scientific research and discovery.

Algorithm Development Intern (Sparse-View Reconstruction for XR) *Beijing, China*
JD Retail, XR Product R&D Department *Jul. 2026 – Sep. 2026*

- Conducted a technical study of sparse-view reconstruction for XR, covering few-view modeling, novel view synthesis, and generative geometric priors.

Honors and Awards

First Prize (Rank 1), Sparse-View Track, Volumetric Video Challenge *Sep. 2026*
The 2nd Volumetric Video Challenge Workshop, SIGGRAPH Asia 2026

Second Prize, The 8th Tsinghua University Artificial Intelligence Challenge (THUAI-8) *May 2025*

Academic Excellence Scholarship, Tsinghua University *2024*

Skills

- **Programming**: Python, C/C++
- **Machine Learning & Tools**: PyTorch, Git, generative models, 3D / 4D reconstruction
- **Languages**: Chinese (native), English (CET-4 647, CET-6 547)